SIMATS ENGINEERING

CO1-ASSESSMENT TOOL 2-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.
(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

QUESTIONS: **Total Marks: 100**

1. Variables and Data Types Analysis (BL4 – Analyze)
Design pseudocode to simulate basic variable handling in R programming:

- Declare variables of different data types (numeric, character, logical)
- Perform arithmetic and logical operations
- Display the results

2. Data Structures Implementation (BL4 – Analyze)
Write pseudocode to demonstrate advanced data structures in R:

- Create and initialize a vector, list, and matrix
- Perform operations such as indexing, updating, and aggregation
- Compare the structures based on usage

3. Matrix and Array Operations (BL4 – Analyze)
Design pseudocode to perform operations on matrices and arrays:

- Create a matrix and a 3D array
- Perform addition, multiplication (matrix), and element-wise operations
- Display the results and analyze differences

4. Data Frame Construction and Manipulation (BL4 – Analyze))

A dataset needs to be organized efficiently.
Write pseudocode to:

- Create a data frame with multiple columns (numeric and character)
- Add, delete, and modify columns
- Perform basic summary operations

5. Functions and Logical Operations (BL4 – Analyze)
Design pseudocode to implement functions and logical vector operations:

- Create a user-defined function with input parameters
- Perform operations on logical vectors
- Return and display computed results

Code of Conduct:

I Hasya Sai Reddy Venati (192420010) +certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	**20%**	**Identifies all inputs, outputs, constraints accurately**	**Minor missing elements**	**Partial understanding**	**Incorrect interpretation**
Algorithm Design	**30%**	**Complete, correct, and logically sound solution**	**Minor logical gaps**	**Partially correct logic**	**Incorrect approach**
Use of Control Structures	**20%**	**Efficient and appropriate use of loops, conditions, functions/modules**	**Minor inefficiencies**	**Limited or basic usage**	**Incorrect or missing constructs**
Pseudocode Structure	**10%**	**Well-structured, clear, consistent format, easy to follow**	**Mostly clear with minor issues**	**Difficult to follow in parts**	**Poorly structured and unclear**
Completeness	**20%**	**Handles all cases; optimized approach**	**Handles most cases; reasonable efficiency**	**Handles basic cases only**	**Incomplete or inefficient**

Q1.
Pseudocode:
num1 <- 20
num2 <- 10
name <- "Student"
passed <- TRUE

addition <- num1 + num2
subtraction <- num1 - num2
multiplication <- num1 * num2
division <- num1 / num2
comparison <- num1 > num2
logical_result <- passed & (num1 > 15)

print(num1)
print(name)
print(passed)
print(addition)
print(subtraction)
print(multiplication)
print(division)
print(comparison)
print(logical_result)

Q2.

```
marks <- c(70, 80, 90, 60, 75)
student <- list(Name="Arun", Age=20, Marks=85)
scores <- matrix(c(70,80,90,60,75,85), nrow=2)

print(marks)
print(marks[2])
marks[2] <- 85
print(marks)
print(sum(marks))
print(mean(marks))

print(student)
print(student$Name)
student$Age <- 21
print(student)

print(scores)
print(scores[1,2])
scores[1,2] <- 95
print(scores)

print("Vector: stores similar values")
print("List: stores different types of values")
print("Matrix: stores values in rows and columns")
```

Q3.

```
m1 <- matrix(c(1,2,3,4), nrow=2)
m2 <- matrix(c(5,6,7,8), nrow=2)

arr <- array(1:8, dim=c(2,2,2))

addition <- m1 + m2
multiplication <- m1 %*% m2
elementwise <- m1 * m2

print(m1)
print(m2)
print(arr)
print(addition)
print(multiplication)
print(elementwise)

print(m1[1,2])
print(arr[1,2,1])
```

Q4.

```
students <- data.frame(
  ID=c(101,102,103,104,105),
  Name=c("Arun","Bala","Charan","Deepak","Elan"),
  Marks=c(80,75,90,65,85),
  Department=c("IT","CSE","ECE","IT","CSE")
)

print(students)

students$Grade <- c("A","B","A","C","A")
print(students)

students$Marks <- students$Marks + 5
print(students)

students$Department <- NULL
print(students)

print(mean(students$Marks))
print(max(students$Marks))
print(min(students$Marks))
print(summary(students))
```

Q5.

```
calculate <- function(a, b) {
 result <- a + b
 return(result)
}

x <- c(10,20,30,40,50)
y <- c(5,25,15,45,35)

result <- calculate(20,30)
```

```
print(result)

greater <- x > 25
print(greater)

selected <- x[x > 25]
print(selected)

logical_result <- x > 25 & y > 20
print(logical_result)

passed <- x >= 30
print(passed)

final <- x[passed]
print(final)
```